

Lec 19

Recap

① Continuous RV X is st its CDF $F_X(x)$

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

$f_X(\cdot)$ as p.d.f (probability density function) of X

② $f_X(x) \geq 0$ a.e

$$\int_{-\infty}^{\infty} f_X(x) dx = 1.$$

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

$$P(X \in B) = \int_{x \in B} f_X(x) dx.$$

Expectation of continuous RV

RHS

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Properties

$$\textcircled{1} E[X] = \int_0^{\infty} P(X > x) dx$$

(for a non-negative RV $\Rightarrow f_X(x) = 0 \forall x < 0$)

Equivalent property for discrete RV

$$E[X] = \sum_{x=0}^{\infty} P(X > x)$$

RHS

$$\int_0^{\infty} P(X > x) dx$$

$$= \int_{x=0}^{\infty} \left[\int_{t=x}^{\infty} f_X(t) dt \right] dx$$

$(x, t) \in [0, \infty)$ st $t \geq x$

$$= \int_{t=0}^{\infty} \left[\int_{x=0}^t dx \right] f_X(t) dt$$

$$= \int_{t=0}^{\infty} t f_X(t) dt$$

$$= E[X] \quad (\text{as } X \text{ is non-negative R.V.})$$

For discrete

$$E[g(X)] = \sum_{x \in E} g(x) P_X(x)$$

$$Y = g(X)$$

$$P_Y(y) = \sum_{x: g(x)=y} P_X(x)$$

For discrete RV X

$$E[g(X)] = \sum_{x \in \mathcal{X}} g(x) P_X(x)$$

$$Y = g(X)$$

$$P_Y(y) = \sum_{x: g(x)=y} P_X(x)$$

R.V.)

② Suppose $Y = g(X)$. st Y is

To show that $E[g(X)] =$

We'll use 1st prop

$$\begin{aligned} E[Y] &= \int_0^\infty y P_Y(y) dy \\ &= \int_0^\infty y P_Y(y) dy \\ &= \int_0^\infty y P_Y(y) dy \end{aligned}$$

② Suppose $Y = g(X)$, st Y is non-negative R.V

To show that $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$

We'll use 1st property.

$$E[Y] = \int_0^{\infty} P(Y > y) dy$$

$$= \int_0^{\infty} P(g(X) > y) dy$$

$$= \int_0^{\infty} \left[\int_{x \in B = \{x: g(x) > y\}} f_X(x) dx \right] dy$$

$$B_1 = \{$$

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R.V

$$\int_{-\infty}^{\infty} f_X(x) dx$$

dy

dy

dy

dy

$$\mathcal{B}_1 = \{(-\infty, x] \mid x \in \mathbb{R}\}$$

$\mathcal{B}_1$ as σ -algebra generated

by $\mathcal{B}_1$

Satisfies
event space
property

$$(a, b) \in \mathcal{B}_1$$

$$[a, b) \in \mathcal{B}_1$$

$$(x, \infty) \in \mathcal{B}_1$$

$$\mathcal{F} = \{\emptyset, \Omega, A, A^c\} \rightarrow \text{generated by } A$$

$$E[Y] = \int_{y=0}^{\infty} \left[\int_{g(x) > y} f_X(x) dx \right] dy.$$

$$\boxed{\begin{array}{l} (x, y) \in \mathbb{R} \times [0, \infty) \\ g(x) > y \end{array}}$$

$$= \int_{x=-\infty}^{\infty} \left[\int_{y=0}^{g(x)} dy \right] f_X(x) dx.$$

$$= \int_{x=-\infty}^{\infty} g(x) f_X(x) dx.$$

$\Rightarrow Y = g(X)$ is non-negative R.V

$$\text{then } E[g(X)] = \int_{x=-\infty}^{\infty} g(x) f_X(x) dx.$$

}

③ For any R.V. X ,

$$E[X] = \int_0^{\infty} P(X > x) dx = \int_{x=0}^{\infty} P(X > x) dx$$

RHS

$$\int_0^{\infty} P(X > x) dx = \int_{x=0}^{\infty} \left[\int_{t=x}^{\infty} f_X(t) dt \right] dx$$

$$\boxed{\begin{matrix} (x, t) \in [0, \infty) \\ t \geq x \end{matrix}}$$

$$= \int_{t=0}^{\infty} \left[\int_{x=0}^t dx \right] f_X(t) dt$$

$$= \int_{t=0}^{\infty} t f_X(t) dt \rightarrow \textcircled{1}$$

$$\int_0^{\infty} P(X < -x) dx$$

$$= \int_{x=0}^{\infty} \left[\int_{t=-\infty}^{-x} f_X(t) dt \right] dx$$

$$(x, t) \in [0, \infty) \times (-\infty, 0]$$

$$t \leq -x \Leftrightarrow x \leq -t$$

$$= \int_{t=-\infty}^0 \left[\int_{x=0}^{-t} dx \right] f_X(t) dt$$

$$= \int_{t=-\infty}^0 -t f_X(t) dt \rightarrow \textcircled{2}$$

$$\textcircled{1} - \textcircled{2} = \int_{t=-\infty}^{\infty} t f_X(t) dt = E[X]$$

For dis

$$E[g(X)] = \int g(x) f_X(x) dx$$

$$f_X(x) = \sum_{x_i} p_i \delta(x - x_i)$$

$$P_Y(y) = \sum_{x: g(x)=y} f_X(x)$$

④ for any R.V, X , $Y = g(X)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Use property 3 to prove (Exercise)

$$\textcircled{5} \quad E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$\textcircled{6} \quad \text{Var}(X) = E[(X - E[X])^2]$$

$$E[X^2] - (E[X])^2 = \int_{-\infty}^{\infty} (x - E[X])^2 f_X(x) dx$$

$$\begin{aligned} &= \int_{-\infty}^{\infty} (x^2 - 2xE[X] + (E[X])^2) f_X(x) dx \\ &= E[X^2] - 2E[X] \int_{-\infty}^{\infty} x f_X(x) dx + (E[X])^2 \int_{-\infty}^{\infty} f_X(x) dx \end{aligned}$$

Proof of 4 is, to show $E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x) dx$

Let $Y = g(x)$. From property 3

$$E[Y] = \int_{y=0}^{\infty} P(Y > y) dy - \int_{y=0}^{\infty} P(Y < -y) dy.$$

$$= \int_{y=0}^{\infty} P(g(x) > y) dy - \int_{y=0}^{\infty} P(g(x) < -y) dy$$

$$= \int_{y=0}^{\infty} \left[\int_{x: g(x) > y} f_x(x) dx \right] dy - \int_{y=0}^{\infty} \left[\int_{x: g(x) < -y} f_x(x) dx \right] dy$$

$$= \int_{x: g(x) > 0} \left[\int_{y=0}^{g(x)} dy \right] f_x(x) dx - \int_{x: g(x) < 0} \left[\int_{y=0}^{-g(x)} dy \right] f_x(x) dx$$

$$= \int_{x: g(x) > 0} g(x) f_x(x) dx + \int_{x: g(x) < 0} g(x) f_x(x) dx.$$

$$= \int_{x=-\infty}^{\infty} g(x) f_x(x) dx. \quad (x: g(x) = 0, g(x) \text{ is anyway zero})$$

Example

$X \sim \text{Uniform}([0, 1])$

$$f_X(x) = \begin{cases} 1 & x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

$$= \int_0^1 x \cdot 1 \cdot dx.$$

$$= \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^1 x^2 \cdot 1 \cdot dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}.$$